ANNEXURE – I

Brief Description of the Design

Digital Leaky Integrate and Fire (LIF) Neuron

Introduction

Spiking Neural Networks (SNNs) represent the third generation of neural networks, drawing direct inspiration from biological brain function. Unlike traditional Artificial Neural Networks (ANNs) that process continuous-valued data in a synchronous, frame-based manner, SNNs are event-driven. They communicate using asynchronous binary "spikes" precisely when a neuron's activation potential reaches a threshold. This bio-mimicry not only offers a more biologically plausible model of computation but also presents significant advantages in power efficiency and temporal data processing, making SNNs ideal for low-power edge computing, robotics, and real-time sensory processing. The most common and computationally efficient model for a spiking neuron is the Leaky Integrate-and-Fire (LIF) model. This model captures the essential dynamics of a biological neuron as listed below.

(1) Integrates incoming signals (spike currents) over time to generate membrane potential.
(2) Gradually leaks its stored membrane potential.
(3) Fires a spike when the membrane potential crosses a predefined threshold.

Workflow

The LIF neuron Integrated Circuit (IC) is a digital hardware realization of a biological neuron that performs temporal integration, leakage, threshold comparison, and spike generation. The IC receives a 16-bit digital input representing the summed contribution of all presynaptic spikes multiplied by their corresponding synaptic weights. This pre-summed input serves as the effective input current to the neuron, capturing the total synaptic excitation or inhibition received from connected neurons. The integrator block combines this input with the neuron's previous membrane potential, which is stored and fed back through a register. This feedback mechanism allows the neuron to retain memory of its past activity, enabling temporal processing across clock cycles.

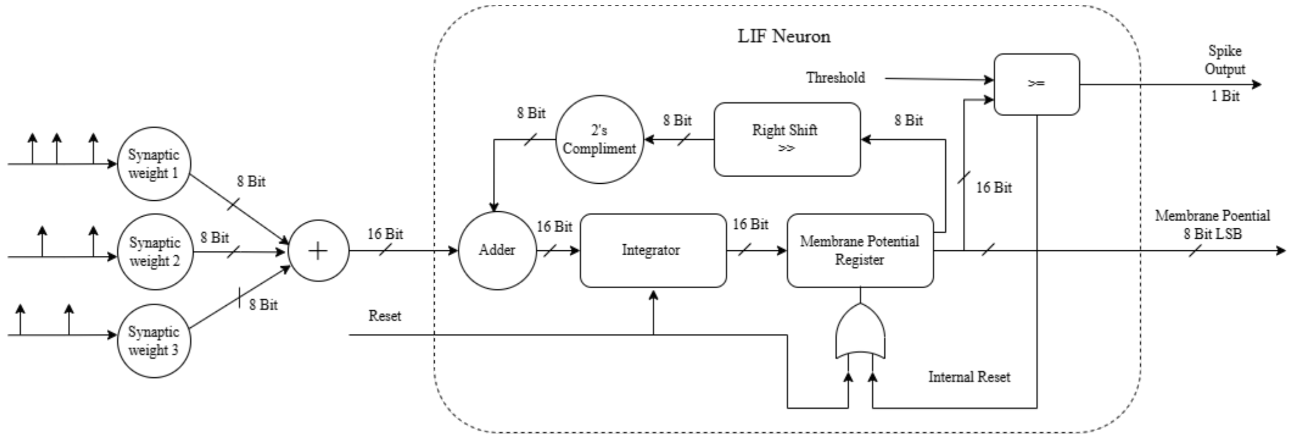

Figure: Work flow of the Design

The updated potential is stored in the membrane potential register, representing the neuron's internal charge state. To emulate biological charge leakage, a right-shift operation is applied to this stored value, effectively scaling it down by a fixed ratio that corresponds to the leakage constant. The result of this right shift is then sent through a 2's complement circuit, which generates its negative equivalent. This negative value is fed back to the adder, allowing it to subtract the leaked portion from the integrated membrane potential. Through this process, the neuron gradually loses potential over time if no new inputs are received, precisely mimicking the natural decay behavior of biological neurons. The resulting membrane potential is then passed to a threshold comparison unit, where it is continuously compared with a preset threshold value. When the membrane potential exceeds or equals this threshold, the neuron generates a digital spike output. Following spike generation, the reset logic discharges the membrane potential to a resting value, typically zero, ensuring the neuron exhibits a refractory period before the next spike can occur.

All functional units including the integrator, right-shift leakage circuit, 2’s complement subtractor, comparator, and reset logic are coordinated by a central clock and control logic that synchronizes operations per clock cycle, ensuring precise timing and deterministic behavior. The outputs of the IC include an 8-bit membrane potential, representing the instantaneous state of the neuron, and a binary spike output, representing the asynchronous event occurrence. These outputs can be cascaded with other neuron ICs to build large-scale spiking neural network (SNN) architectures, enabling efficient neuromorphic computing with real-time digital processing.

Table: Technical Specifications for Leaky Integrate and Fire (LIF) Neuron architecture

Technology node	180nm SCL PDK
Total Pins	48
Metal Layers	4
Power Supply	1.8V
Clock Frequency	100 MHz
Input Data Width	16 bits
Output Data Width	8 bits
Size of chip	2mm x 2mm

Application

This digital LIF neuron IC is a high-performance, low-level component for building large-scale, custom SNN hardware. Its primary applications lie in neuromorphic computing, where it can be arrayed to create complex, multi-layer networks. Key domains include real-time sensory processing, especially with event-based sensors like DVS for high-speed object tracking and gesture recognition. It is also ideal for low-power edge AI in battery-powered devices (IoT, wearables, drones) that need on-board intelligence without the high power cost of traditional ANNs. Furthermore, it serves robotics and autonomous systems by enabling rapid, event-driven control for tasks like obstacle avoidance and navigation, and provides a platform for computational neuroscience to simulate large biological networks and test hypotheses about brain function.